

Non-hierarchical group testing algorithms and group viral loads

Pranta Das
University of Nebraska–Lincoln
Department of Statistics

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Joint work with
Christopher R. Bilder at University of Nebraska-Lincoln,
Joshua M. Tebbs at University of South Carolina, and
Christopher S. McMahan at Clemson University

What is group testing?

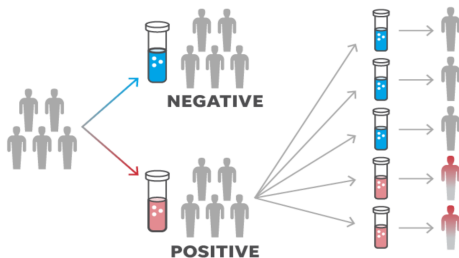
- Specimens from different individuals are combined
- Tested as a single specimen for a pathogen
 - Group tests negative: All individuals in it are declared negative
 - Group tests positive: Algorithms to determine which individuals are positive/negative
- **Key aspect:** Binary responses
- Also known as “pooled testing”
- Idea first introduced by Dorfman (*Annals of Mathematical Statistics*, 1943)

Importance

- Decrease number of tests
 - Increase in testing capacity
 - Works when disease prevalence is low
- A recent instance: COVID-19 pandemic
 - News media outlets: *The New York Times*, *The Washington Post*, ABC News, PBS NewsHour, USA Today, ...
 - Over 1 GB of papers published during the first 2 years of the pandemic!
 - Bilder et al. (*Statistics in Medicine*, 2021)
 - Hogan et al. (*Journal of the American Medical Association*, 2020)
 - Nikolopoulos et al. (*IEEE BITS the Information Theory Magazine*, 2021)
- Widely used elsewhere: Blood donation screening, computer network assessments, ...

Algorithms

- Hierarchical algorithm
 - Dorfman testing

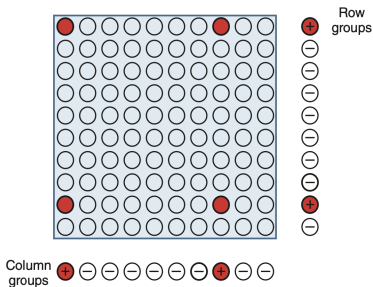


SOURCE USA TODAY research
Karl Gelles/USA TODAY

- Three stages and higher are possible
- Individuals can be tested once in a stage

Algorithms

- Non-hierarchical algorithm
 - Array testing



- **Key aspect:** Test in multiple groups during first stage to reduce the number of retests in a second stage

Purpose

- During COVID-19 pandemic
 - Two new non-hierarchical algorithms were developed (Shental et al. 2020, Ghosh et al. 2021)
 - Use viral load responses rather than binary responses
 - Use linear model to predict positive/negative
 - To avoid retesting as much as possible
- Purpose
 - Examine use of viral load response and linear model prediction with array testing
 - Fair comparisons with other standard algorithms

Pooling matrix

- A 3×3 array

	Column1	Column 2	Column 3
Row 1	1	2	3
Row 2	4	5	6
Row 3	7	8	9

- Alternative form

Groups	Specimens									Regular Array
	1	2	3	4	5	6	7	8	9	
1	1	1	1	0	0	0	0	0	0	Row 1
2	0	0	0	1	1	1	0	0	0	Row 2
3	0	0	0	0	0	0	1	1	1	Row 3
4	1	0	0	1	0	0	1	0	0	Col. 1
5	0	1	0	0	1	0	0	1	0	Col. 2
6	0	0	1	0	0	1	0	0	1	Col. 3

- Pooling matrix: $M_{6 \times 9}$ = matrix of 0's and 1's

Group testing with viral load data

- Define
 - $\mathbf{x}$ as a vector of true individual viral loads; not observable
 - $\mathbf{Y}$ as a vector of group viral loads; observable
 - $\mathbf{Y} = \mathbf{M}\mathbf{x}$
 - Looks like a linear model: $E(\mathbf{Y}) = \mathbf{X}\beta$
- Fit model to estimate $\mathbf{x}$
 - Estimated viral load equal 0: Specimen has no virus
 - Estimated viral load greater than 0: Specimen has virus
 - Estimates of $\mathbf{x}$ are used to predict positive/negative status of specimens
 - # of columns of $\mathbf{M}$ (# of specimens) > # of rows of $\mathbf{M}$ (# of groups)
 - Non-negative LASSO

Predictions of x

- Estimated viral loads greater than or equal to a threshold declare positive
- Otherwise declare negative
- Retest specimens with estimated viral loads below a threshold but greater than 0

Methods

- Algorithms investigated
 - Array with non-negative LASSO (without retest)
 - Array with non-negative LASSO (with retest)
 - Array testing
 - Dorfman testing

Measures

- Expected number of tests per individual
 - Let T be the number of tests for an initial set of I individuals
 - For Dorfman, I represents stage 1 group size
 - For array, I represents number of individuals in an array
 - $E(T)/I$ is the expected number of tests per individual
 - $E(T)/I = 1$ when testing each individual separately
 - Want $E(T)/I < 1$ for group testing
 - “Best” algorithm has the lowest value
- Positive percentage agreement (PPA)
 - Probability of declaring positive given the individual is truly positive
- Closed form expressions for $E(T)/I$ and PPA
 - Do not exist when working with viral loads
 - Use Monte Carlo simulation

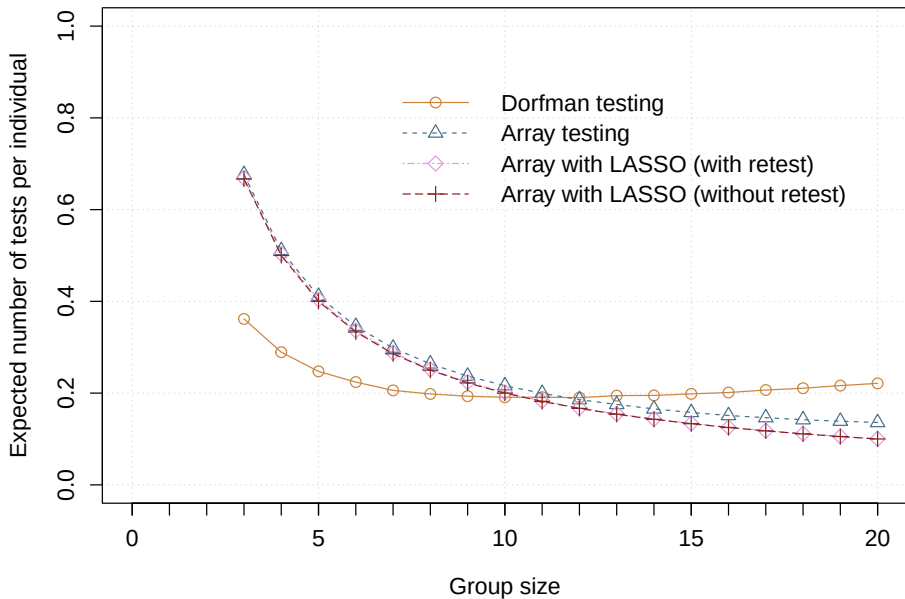
Sketch of data simulation

- Simulate true status using Bernoulli(p)
 - p is disease prevalence
- Simulate the test outcome from an RT-qPCR assay
 - Produce cycle threshold (CT) value as a test response
 - Many papers looked at these tests
 - Draw samples from Hirschhorn et al. (2021)
- If all specimens are negative in a group
 - Group viral load is 0
 - Otherwise
 - Group CT: $\min(\text{Positive specimens CT}) + \log_2(\text{Group Size}) + \epsilon; \epsilon \sim N(0, \sigma^2)$ (Tan et al. 2020)
 - Convert CT value to viral load using expression given by Arnout et al. (2021)
- Upon retest
 - If true status negative, viral load equal to 0
 - Otherwise, Individual CT: $\text{True CT} + \epsilon; \epsilon \sim N(0, \sigma^2)$

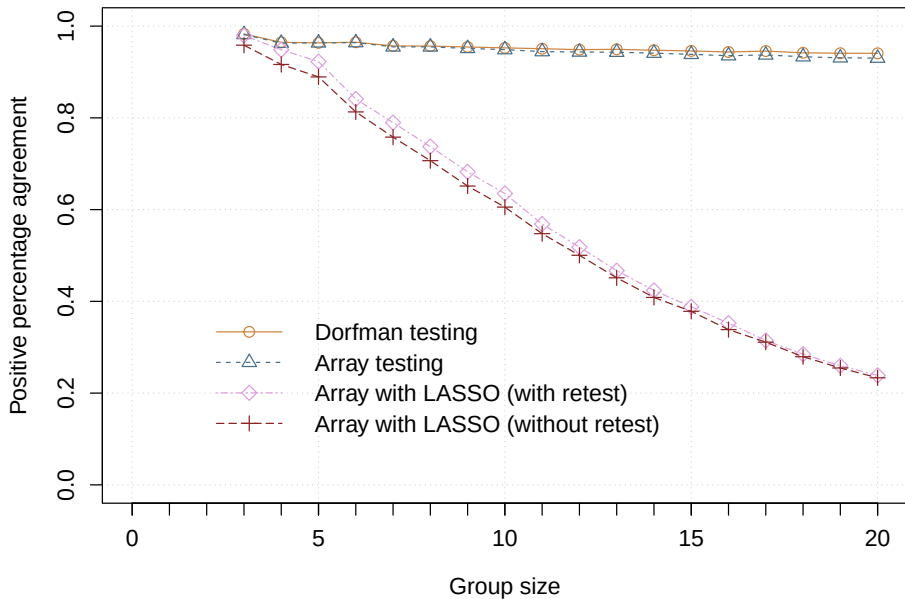
Steps

- ➊ Repeat the data simulation process 10,000 times
- ➋ Observe number of tests, number of tested positives, and true positives each time
- ➌ Estimate $E(T)/I$ and PPA for each algorithm
 - $p = 0.01$ and 0.10
 - **Group size:** 3 to 20
 - $\sigma = 0.1$ and 0.3
 - Threshold: Viral load at $CT = 31.5$
- ➍ Compare algorithms by estimated $E(T)/I$ and PPA

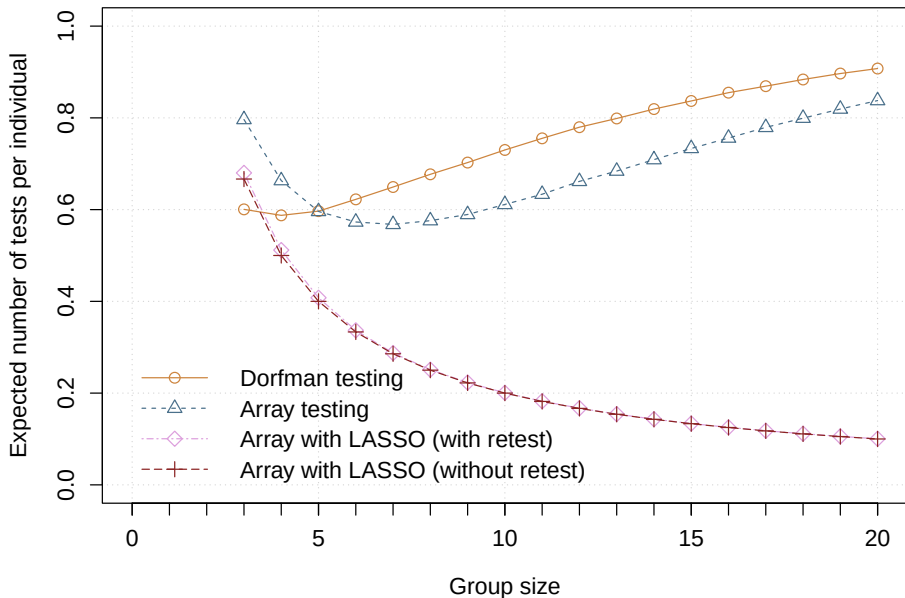
$p = 0.01$ and $\sigma = 0.1$



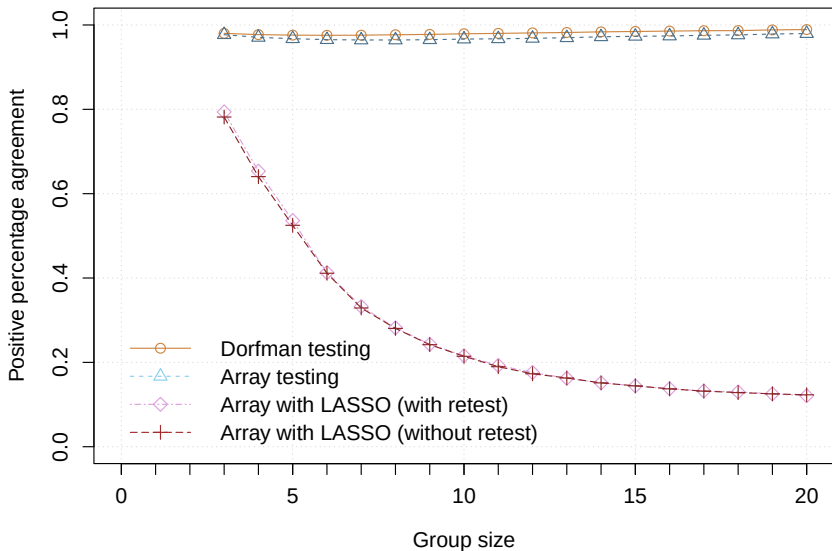
$\rho = 0.01$ and $\sigma = 0.1$



$p = 0.10$ and $\sigma = 0.1$



$\rho = 0.10$ and $\sigma = 0.1$



• Same relationship between algorithms with $\sigma = 0.3$

Conclusion

- LASSO approach
 - Less expected number of tests per individual
 - Low accuracy
- Work continues to improve it

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